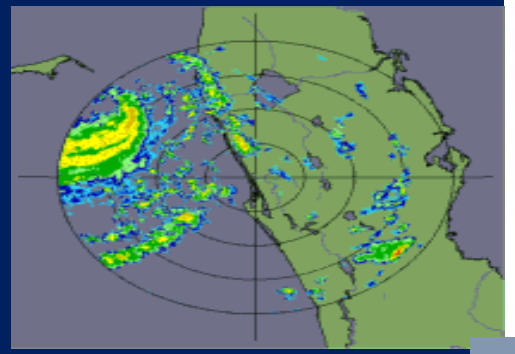
USPCAS-W
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HYDROINFORMATICS

GIS Data Sources and Acquisition

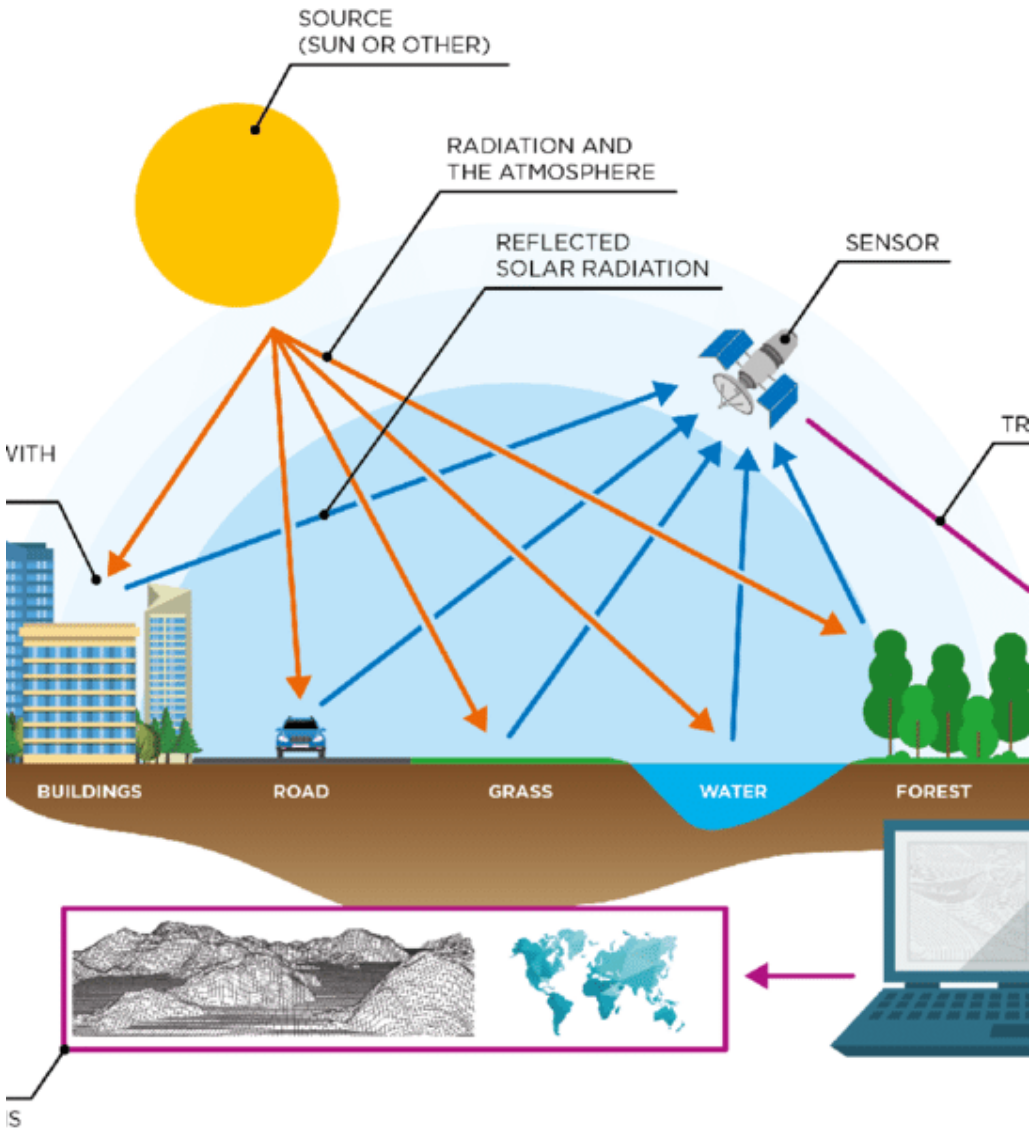
By: Dr. Arjumand Zaidi
Senior Research Fellow






Fall 2026
US Pakistan Center for Advanced Studies in Water
MUET, Jamshoro

Let's kickstart our brains on this topic.



- What are some common on-the-ground methods for data collection?
- What are the primary sources of satellite data for Earth observation?



Learning Objectives

1. To discuss data acquisition challenges.
2. To learn about free satellite data sources.

Data Collection Methods

- **Secondary Data Collection**
  - existing data
  - already published/available
  - Someone else collected data
- **Primary Data Collection**
  - from scratch
  - You collected or assigned someone to collect

Field Measurements

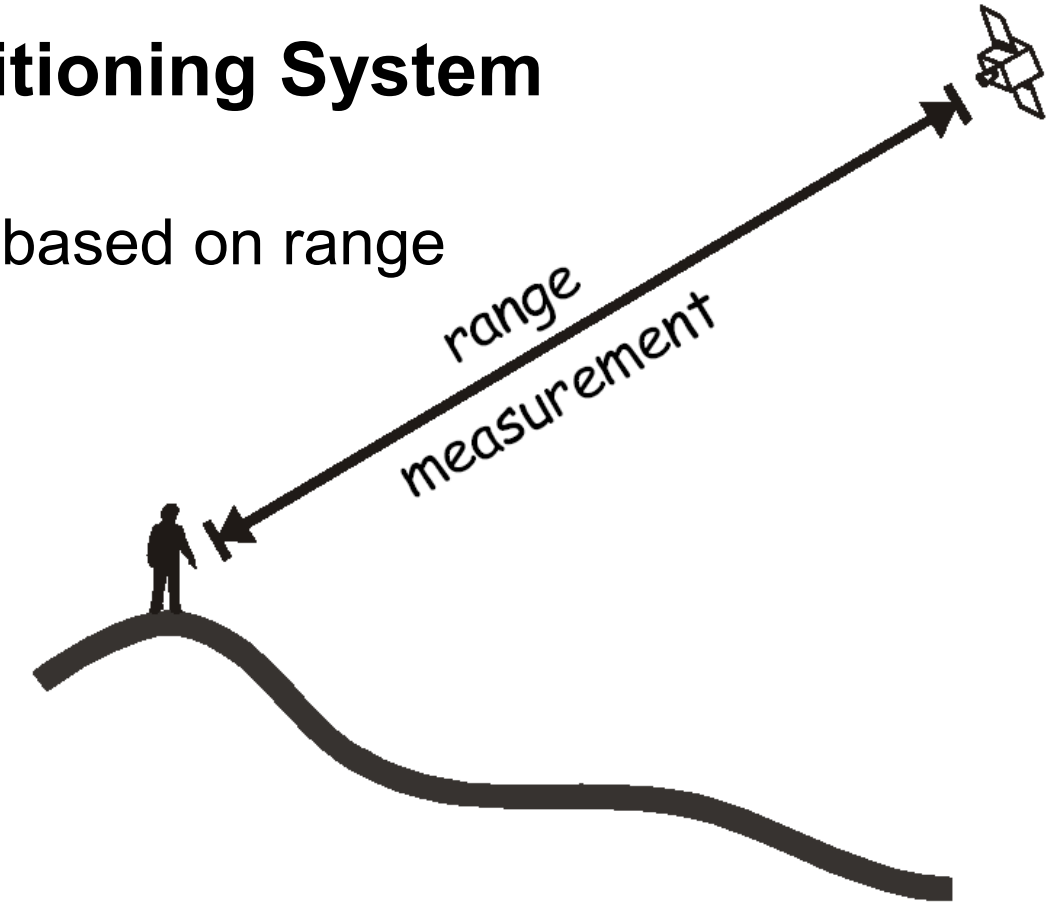




Coordinate Surveying

GPS: Global Positioning System

Position is estimated based on range measurements.



Range = speed of light x travel time
Range = c(t/2)

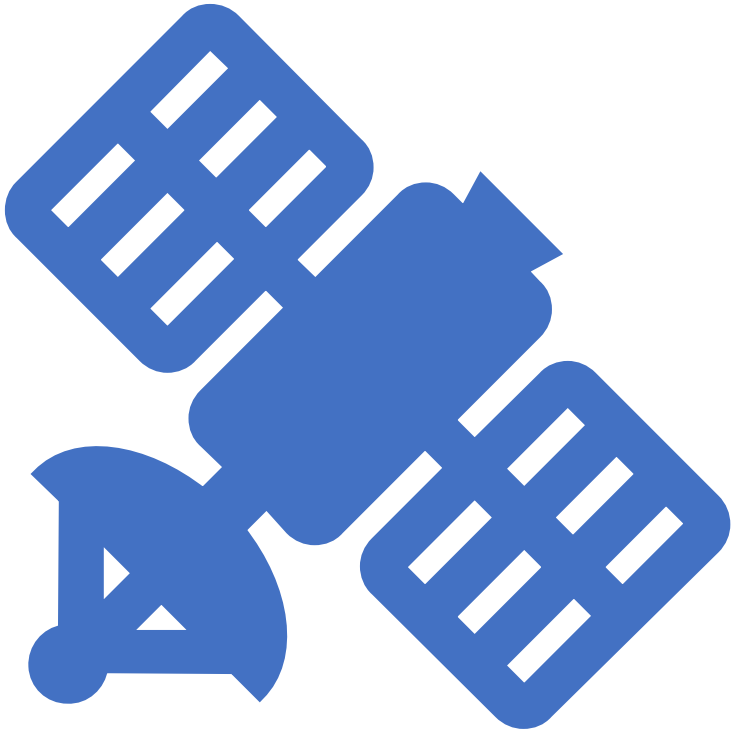
(c =299,792,458 meters per second, t = round trip time of a signal)

Remote Sensing: measuring electromagnetic energy reflected or emitted from objects – airborne or satellite-based instruments





satellites vary by: orbit, altitude, resolution, revisit frequency, revisit variability (steering) capability, width of swath, image size, stereo capability, wavelengths collected, other sensors, customer delivery time, etc.



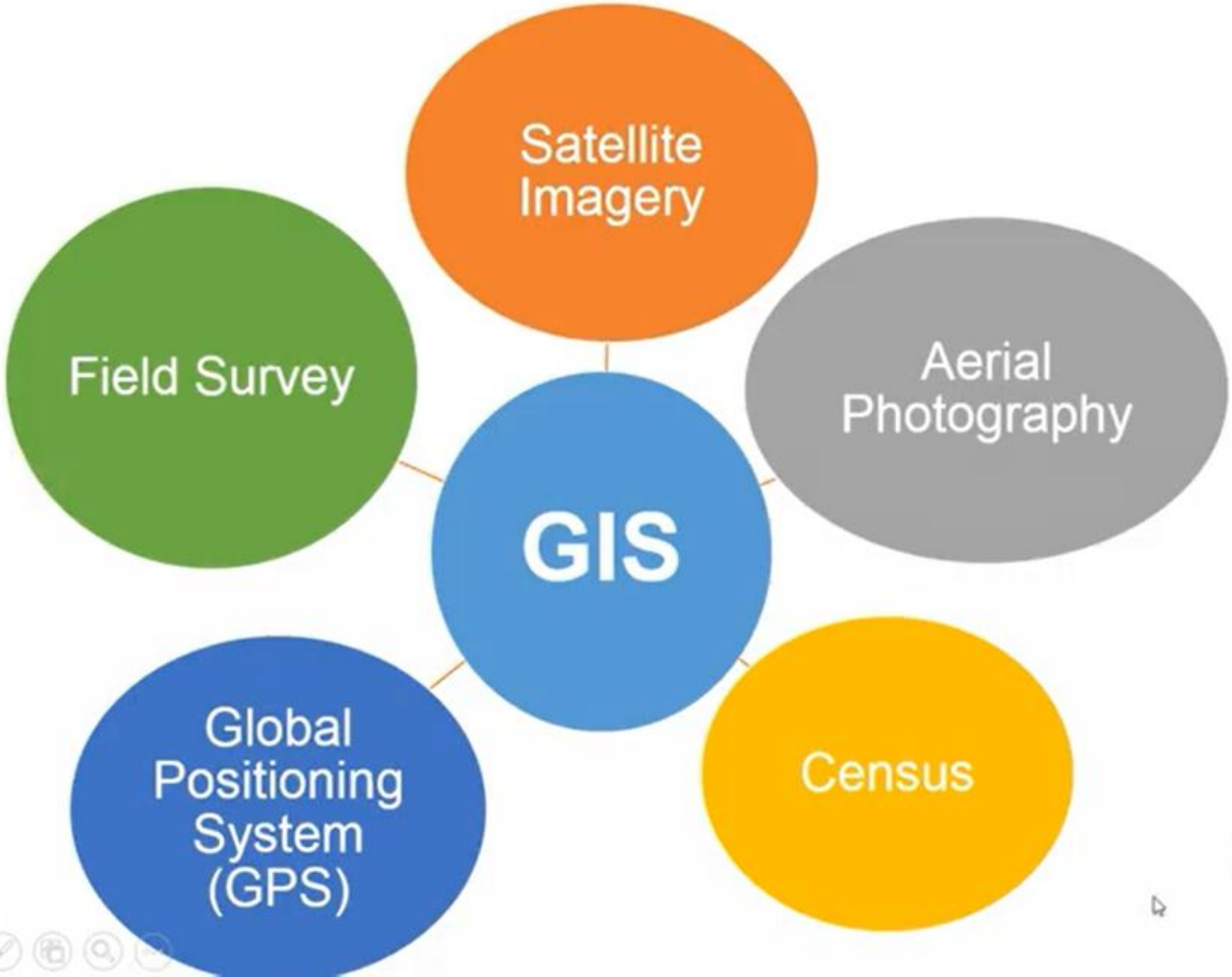
Data Collection Protocols

- Employ a reliable sampling method for representative data.
- Standardize procedures and employ quality control to enhance data reliability and consistency.
- Minimize bias in primary data collection.
- Obtain consent and safeguard participants' confidentiality and rights.

Data use protocols

- Credit secondary data sources through proper citation.
- Ensure data is obtained and used ethically.
  - Adhere to pertinent laws/regulations to ensure legal conformity, safeguard intellectual property rights, and uphold ethical standards.
- Assess the quality and reliability of data before utilization.

Sources of GIS Data



GIS Data Sources

Maps and Drawings

- Scanning or digitizing

Aerial/Satellite Images

- Image interpretation for features extraction
- Digital terrain model) to create digital orthos

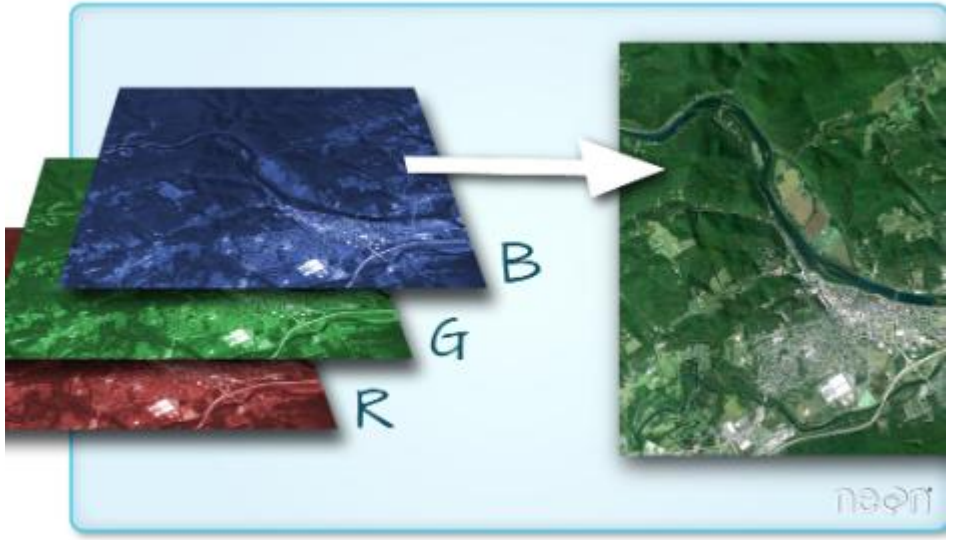
CAD Databases

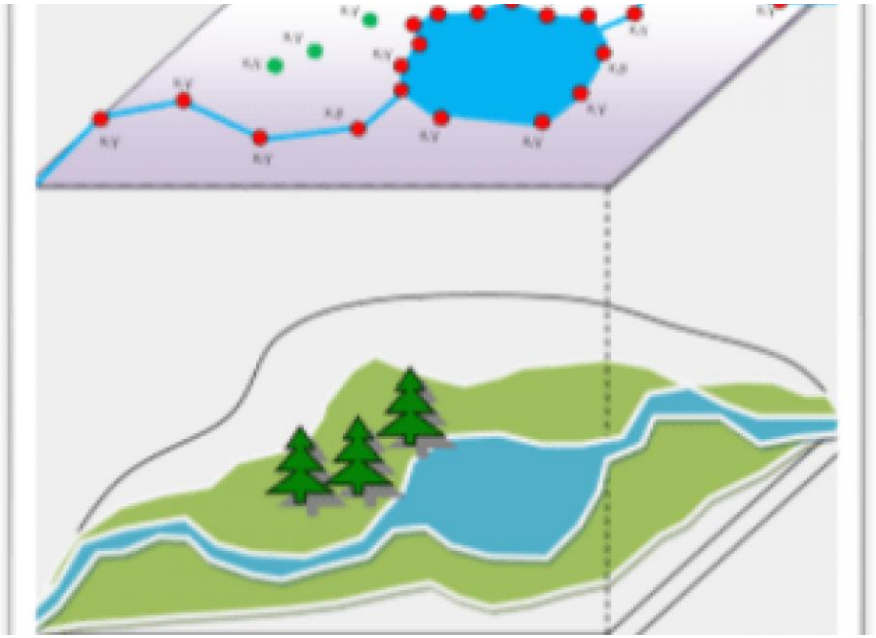
GIS Databases

Attribute Databases

Records and Documents

Sources of Vector/Raster Data







Steer decisions and results with your data

Data is critical to decision making. Whether you are buying a house, wondering about road construction, or opening a new business, trustworthy information is essential. Use Esri's SaaS open data capabilities to provide your community with the authoritative data they need to make better decisions and accomplish their goals.

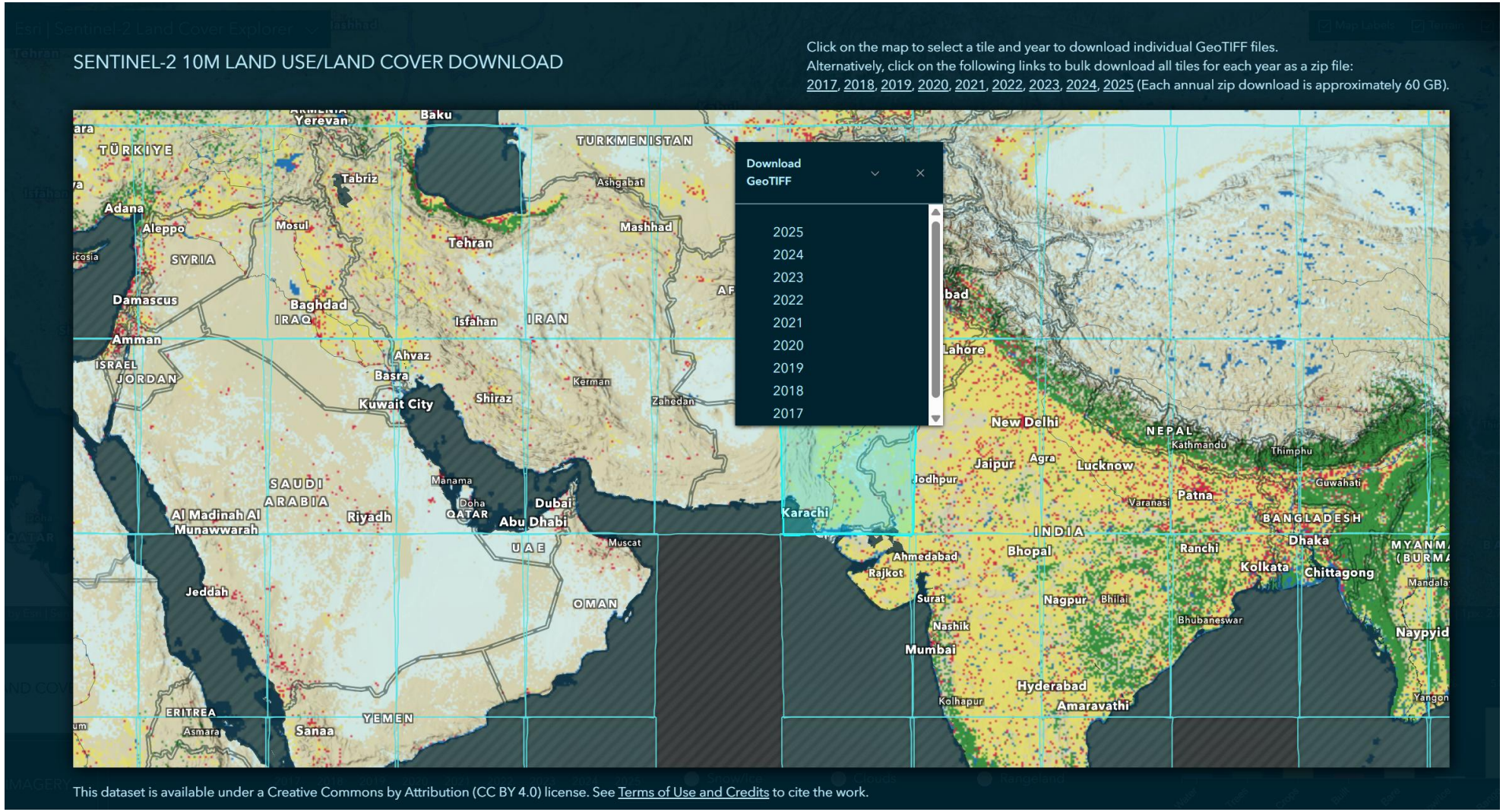
https://www.esri.com/en-us/arcgis/products/arcgis-open-data

Landcover Explorer

https://livingatlas.arcgis.com/landcoverexplorer/



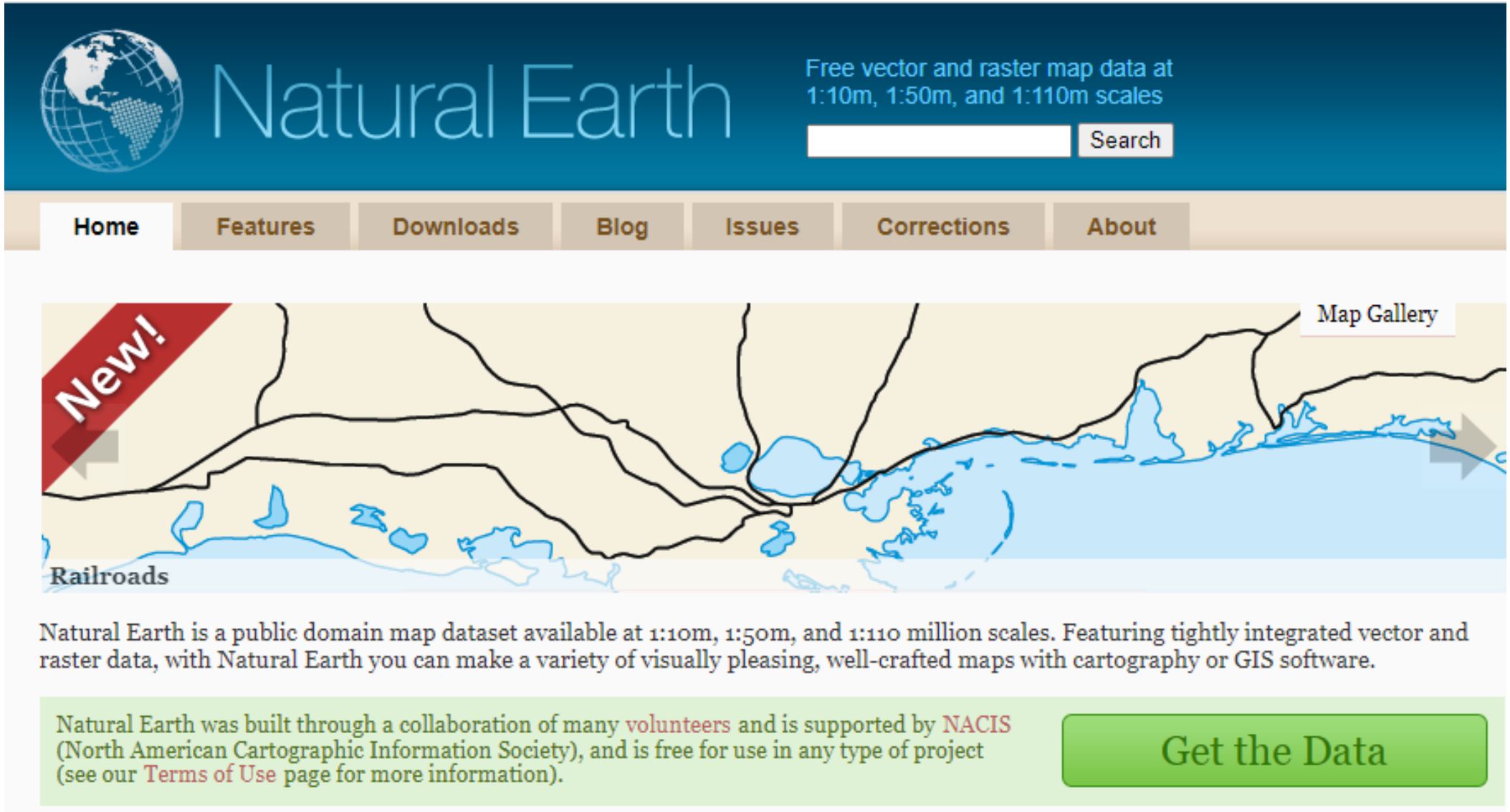
ESRI LULC 2017 - 2025



https://livingatlas.arcgis.com/landcoverexplorer/#mapCenter=70.32492%2C30.11343%2C5.88&mode=step&timeExtent=2017%2C2025&year=2025&downloadMode=true

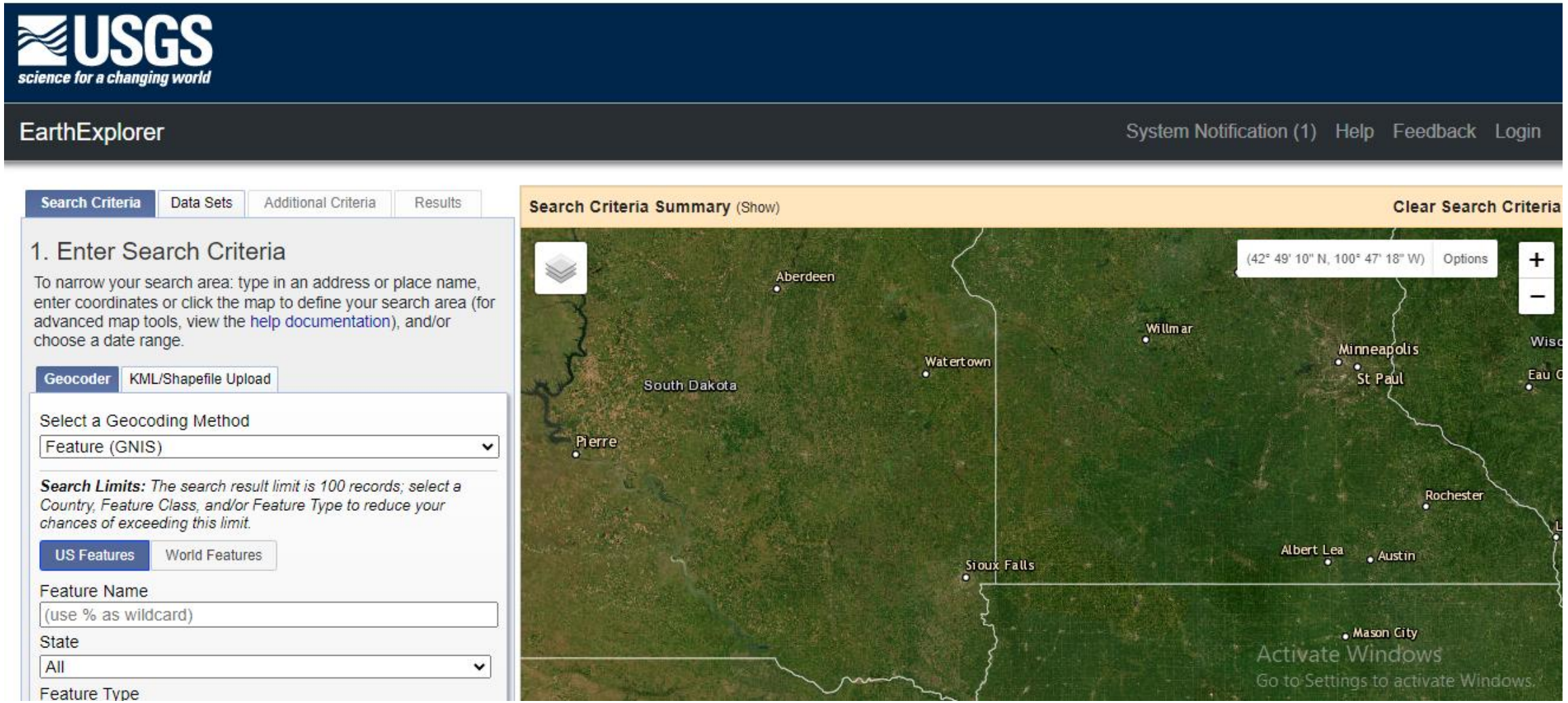
Natural Earth Data

Link: https://www.naturalearthdata.com/

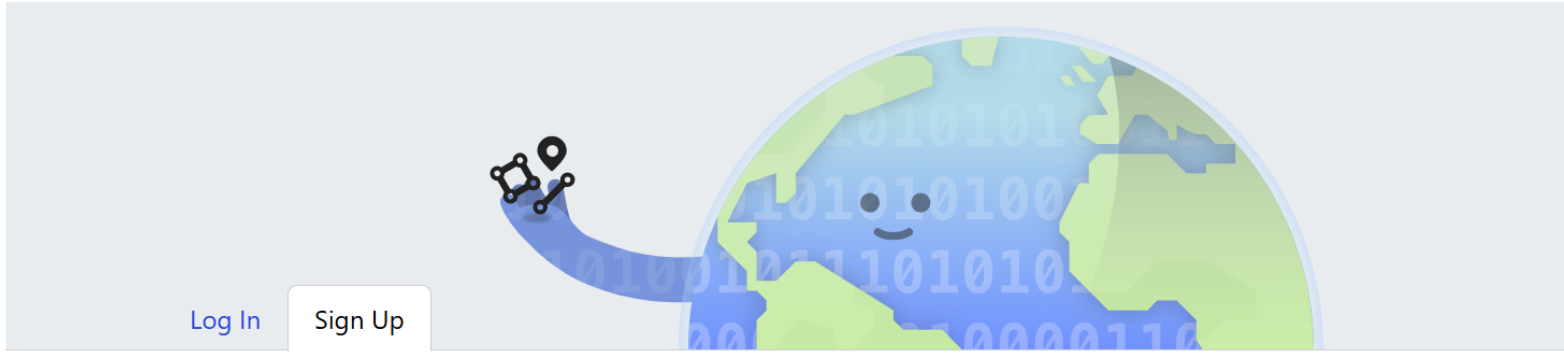


USGS Earth Explorer

Link: https://earthexplorer.usgs.gov/




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AN OPEN SOURCE MAP PROJECT
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Map Data
Public GPS Traces
200 km
https://www.openstreetmap.org/#map=6/30.67/69.36



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Display Name

Your publicly displayed username. You can change this later in the preferences.

Password

Confirm Password

Verify you're a human

Success!

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Sign Up

Another Option for Downloading OSM is from GeoFabrik Downloads

https://www.geofabrik.de/data/download.html



Our Download Server

Our download server at https://download.geofabrik.de has excerpts and derived data from the OpenStreetMap dataset available for free download. Most of these files are updated every day – any change you upload to OpenStreetMap should be on our download server the next day.

Geographic Regions

Data on the download server is organised by region. The root directory contains files that have a whole continent's data in them, and for all continents except Antartica there are subdirectories in which you find individual files for various countries. Some countries again have their own subdirectories with data for administrative subdivisions. For Germany, all "Bundesländer" are available as separate files, and for England we have all individual counties. For France, we have all (historic) Departements. The file for Germany's Baden-Württemberg region is, for example, at

```
https://download.geofabrik.de/europe/germany/baden-wuerttemberg-latest.osm.pbf
```

Please let us know if you are interested in other regional excerpts that we do not offer right now – we might be able to add them with little effort.

Raw data (OSM source data)

Files with the `.osm.pbf` suffix are OpenStreetMap raw data files in the PBF format. They are the common format for OpenStreetMap raw data and need few memory for storage. Daily diff updates (`.osc.gz` , Gzip compressed OSM XML) are available on our download server as well. They contain the changes of one day. The PBF files can be updated with them.

The Bzip2 compressed XML files we used to have on the download server are now discontinued since almost all software supports .osm.pbf. If you still need bz2 files, you can create them yourself, or order them commercially from us.

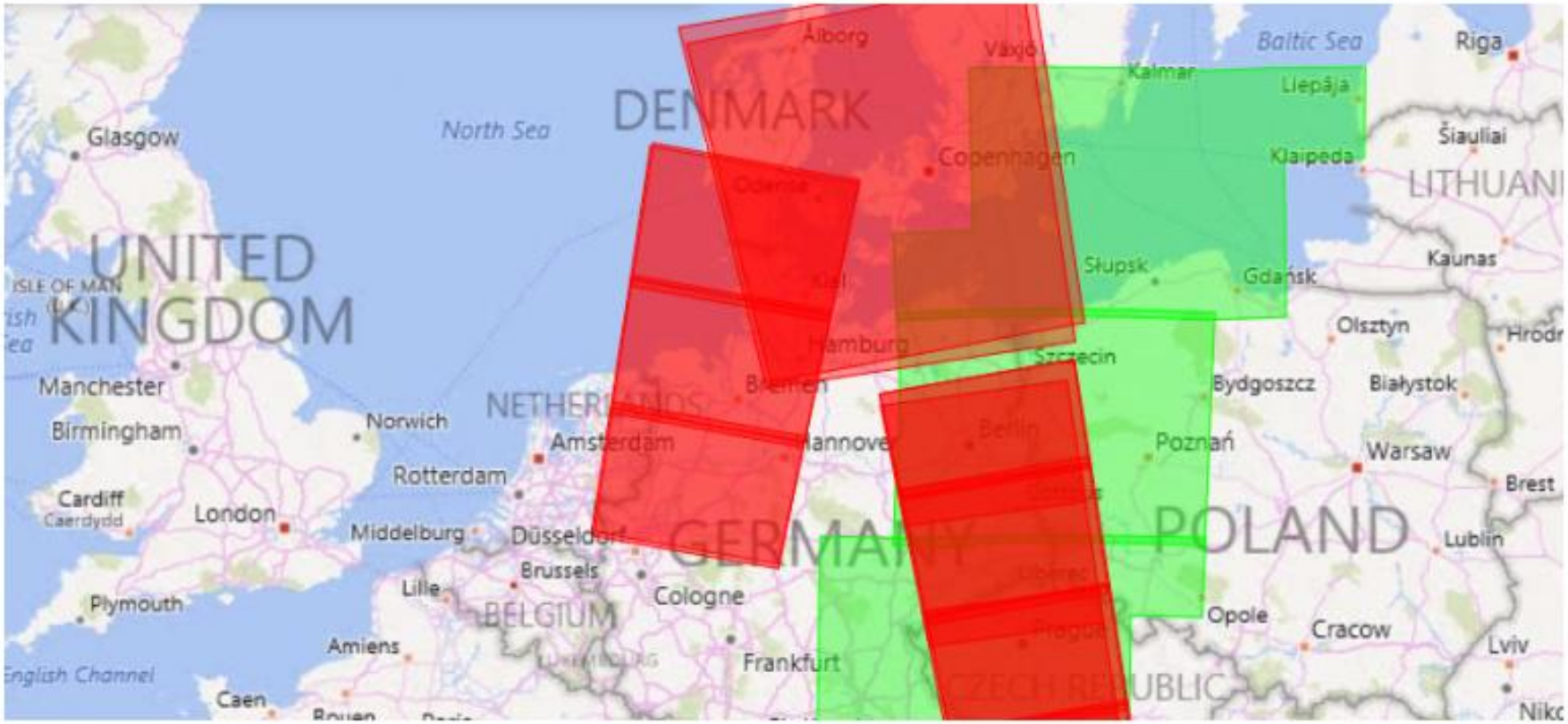
You need special software to process OSM raw data. It is usually command line and server software for Linux. Common tools to process OpenStreetMap raw data are:

Sentinel Scientific Data Hub.

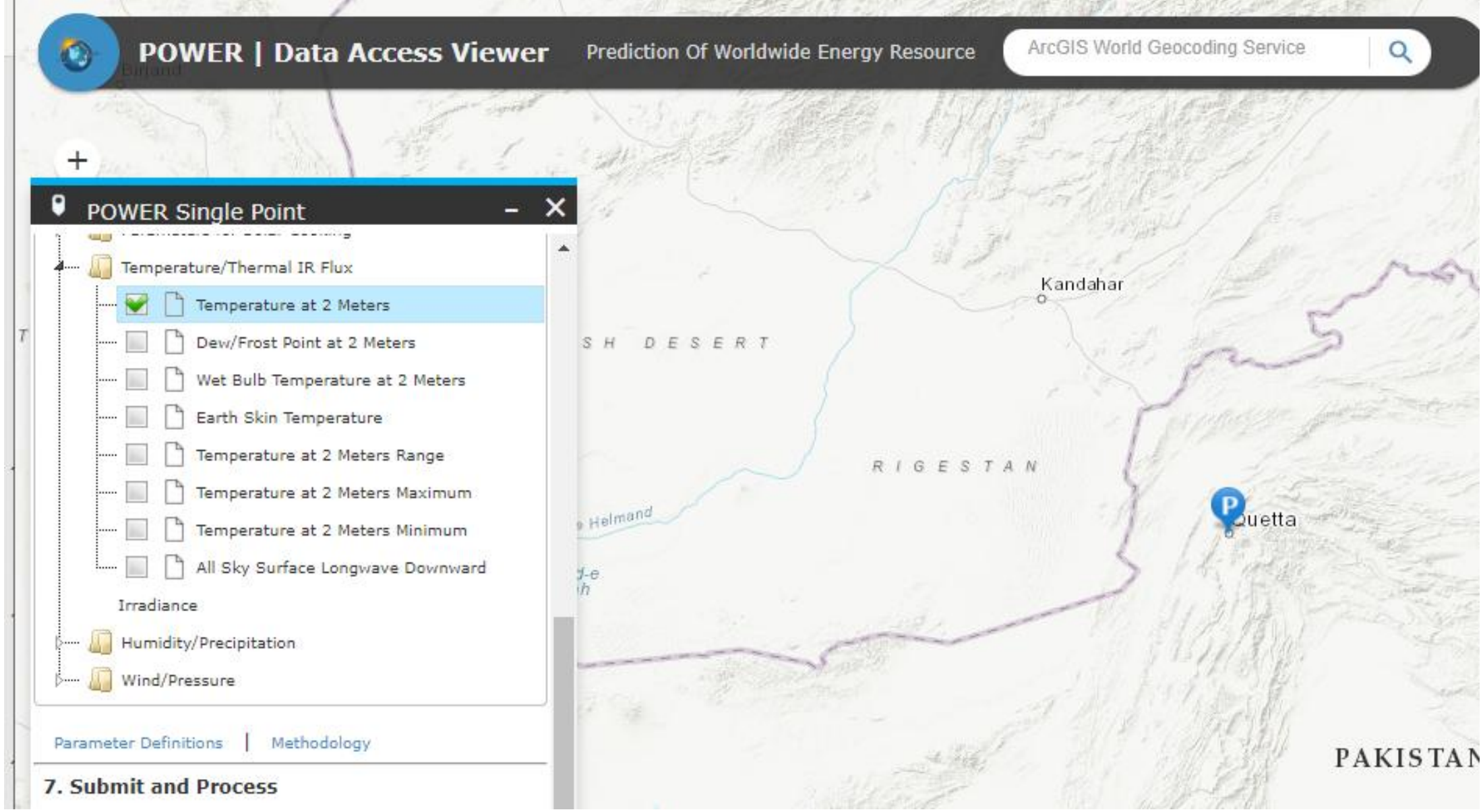
https://gisgeography.com/how-to-download-sentinel-satellite-data/

How to Download Free Sentinel Satellite Data

By: GISGeography • Last Updated: October 1, 2023

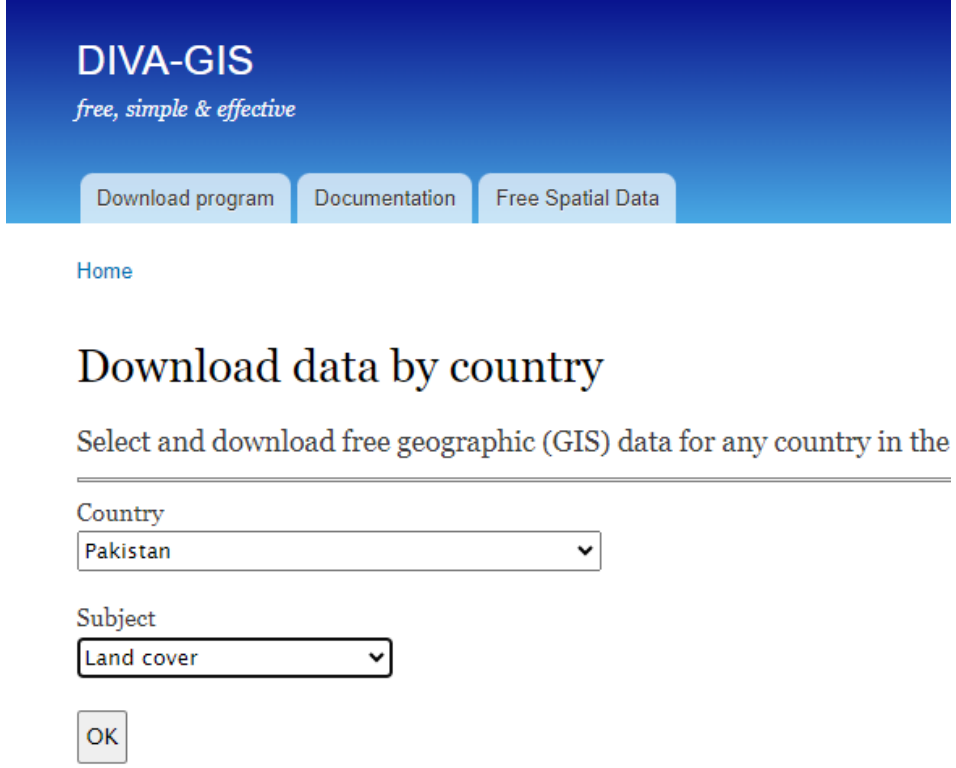


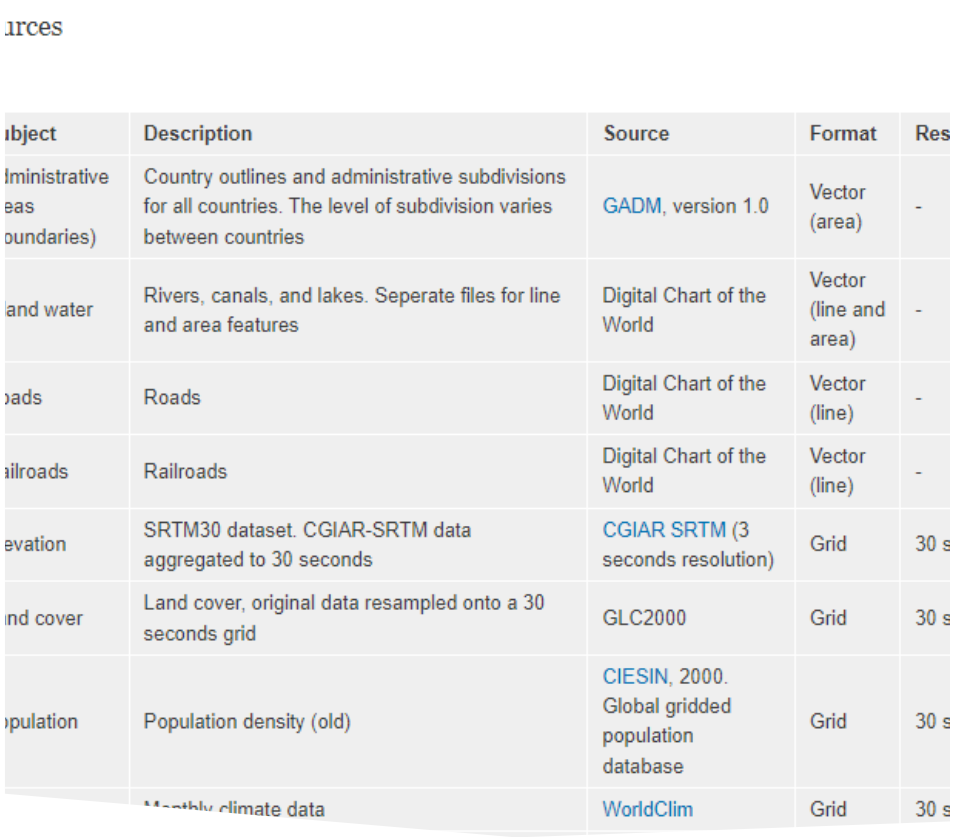
Download Sentinel-1 and Sentinel-2 Satellite Data



Weather Data

Link: https://power.larc.nasa.gov/data-access-viewer/





ıbject	Description	Source	Format	Res
lministrative eas oundaries)	Country outlines and administrative subdivisions for all countries. The level of subdivision varies between countries	GADM, version 1.0	Vector (area)	-
and water	Rivers, canals, and lakes. Seperate files for line and area features	Digital Chart of the World	Vector (line and area)	-
oads	Roads	Digital Chart of the World	Vector (line)	-
ailroads	Railroads	Digital Chart of the World	Vector (line)	-
evation	SRTM30 dataset. CGIAR-SRTM data aggregated to 30 seconds	CGIAR SRTM (3 seconds resolution)	Grid	30 s
nd cover	Land cover, original data resampled onto a 30 seconds grid	GLC2000	Grid	30 s
opulation	Population density (old)	CIESIN, 2000. Global gridded population database	Grid	30 s
	Monthly climate data	WorldClim	Grid	30 s

Diva-GIS:

- **Download Data by Country**

https://www.diva-gis.org/gdata

Diva GIS

Download Data by Country

https://diva-gis.org/data.html

DIVA-GIS

Data

« Documentation :: Contents :: Climate data »

Data

Select and download free geographic (GIS) data for any country in the world
 GIS data download

Country
Pakistan

Subject
Administrative areas

Format

Download **PAK_adm.zip**

Sources

Subject	Description	Source	Format	Resolution
Administrative areas (boundaries)	Country outlines and administrative subdivisions for all countries. The level of subdivisions that is available varies between countries	**GADM**	Vector (area)	-
Inland water	Rivers, canals, and lakes. Seperate files for line and area features	Digital Chart of the World	Vector (line and area)	-
Roads	Roads	Digital Chart of the World	Vector (line)	-
Railroads	Railroads	Digital Chart of the World	Vector (line)	-
Elevation	SRTM30 dataset. CGIAR-SRTM data aggregated to 30 seconds	**CGIAR SRTM** (3 seconds resolution)	Grid	30 seconds
Land cover	Land cover, original data resampled onto a 30 seconds grid	GLC2000	Grid	30 seconds
Population	Population density (old)	**CIESIN**, 2000. Global gridded population database	Grid	30 seconds
Climate	Monthly climate data	**WorldClim**	Grid	30 seconds
Gazetteer	A gazetteer is a list of place names and their coordinates. The files you can download here are for use in DIVA for automatic georeferencing (to assign coordinates to places). The files should be placed in the \\gazet directory. This is very old now. There area better services for this.	DBF	-	

Formats

The files have been compressed and grouped in ZIP files. You can use programs such as 7-zip, PKZIP or StuffIt to decompress the files.

Vector data are stored as ESRI shapefiles Grid (raster) data are stored as DIVA gridfiles

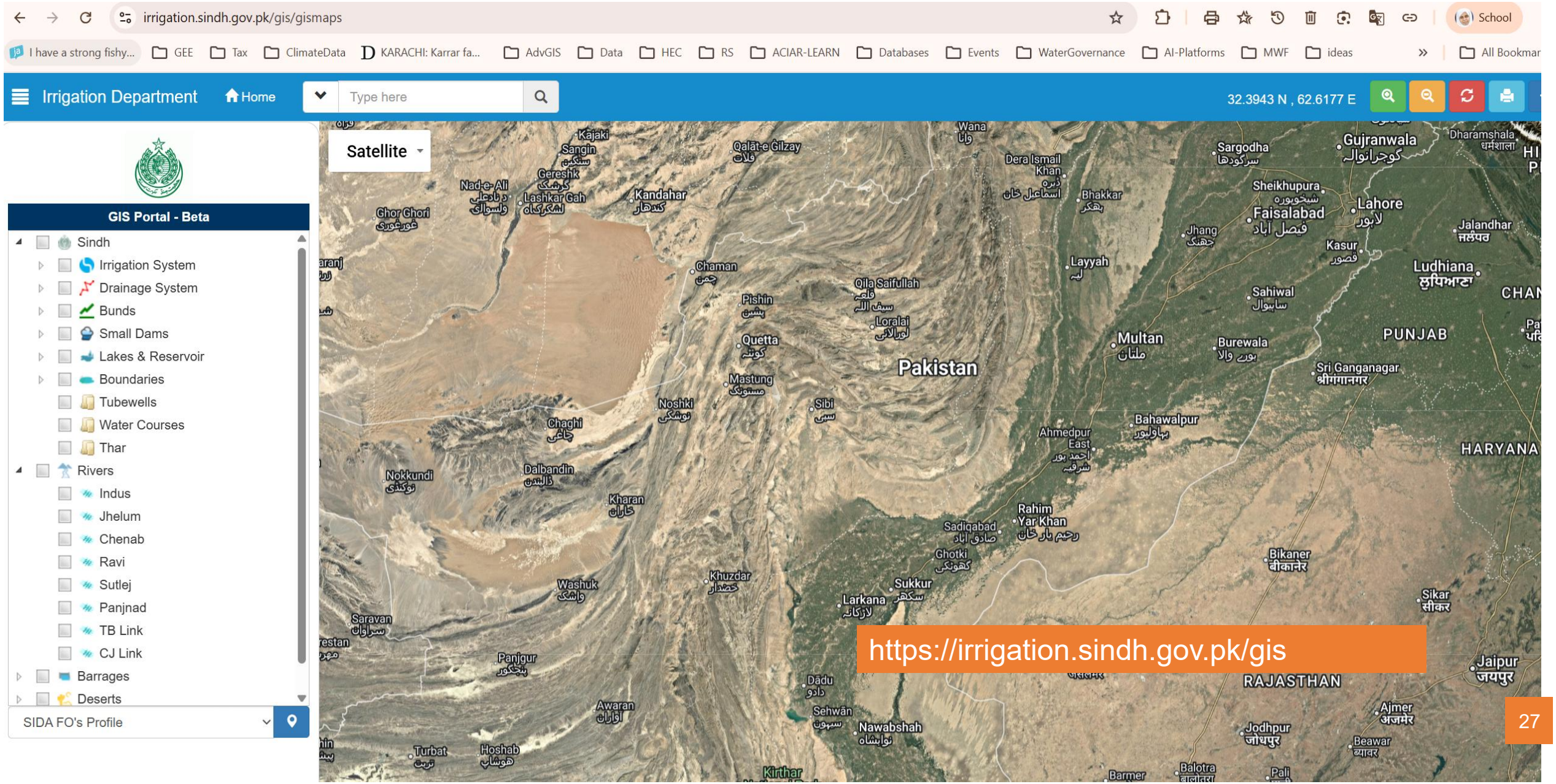
FAO SOILS PORTAL

1 km resolution

https://www.fao.org/soils-portal/data-hub/soil-maps-and-databases/harmonized-world-soil-database-v20/en/



Sindh Irrigation GIS Portal



IRRIGATION DEPARTMENT

Home About Team Attached Bodies Sindh Water Policy Water E-Library Events Contact

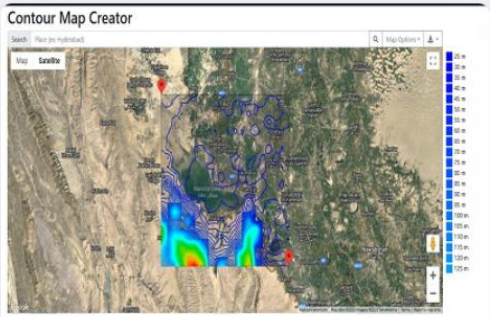
CLOUD APPS GALLERY

Experimental



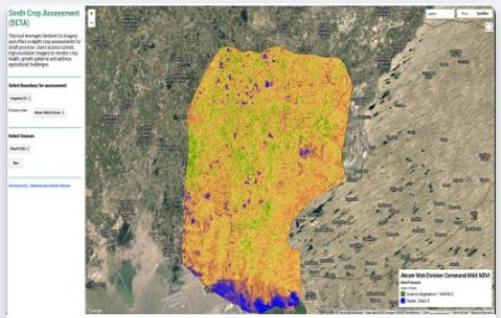
Irrigation GIS Portal

The GIS Portal offers comprehensive geospatial data, facilitating informed decision-making for efficient water resource management.



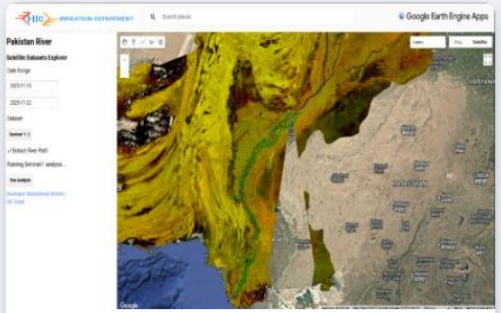
Contour Mapping

This app allows efficient visualization and analysis of land elevations, aiding in planning and water evacuation.



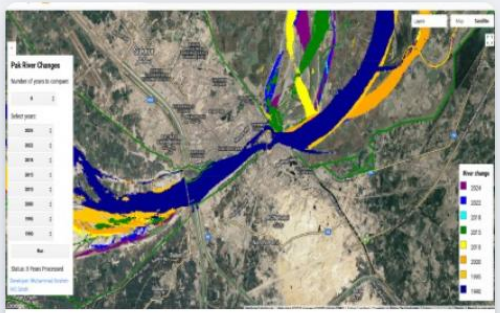
Sindh Crop Assessment

This tool leverages Sentinel-2A imagery and offers in-depth crop assessments for Sindh province.



Pak River Path

This tool leverages Sentinel-2A and Landsat imagery and offers in-depth Pak rivers .



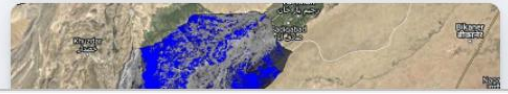
Pak River Path

This tool leverages Landsat imagery and offers changes in rivers path .



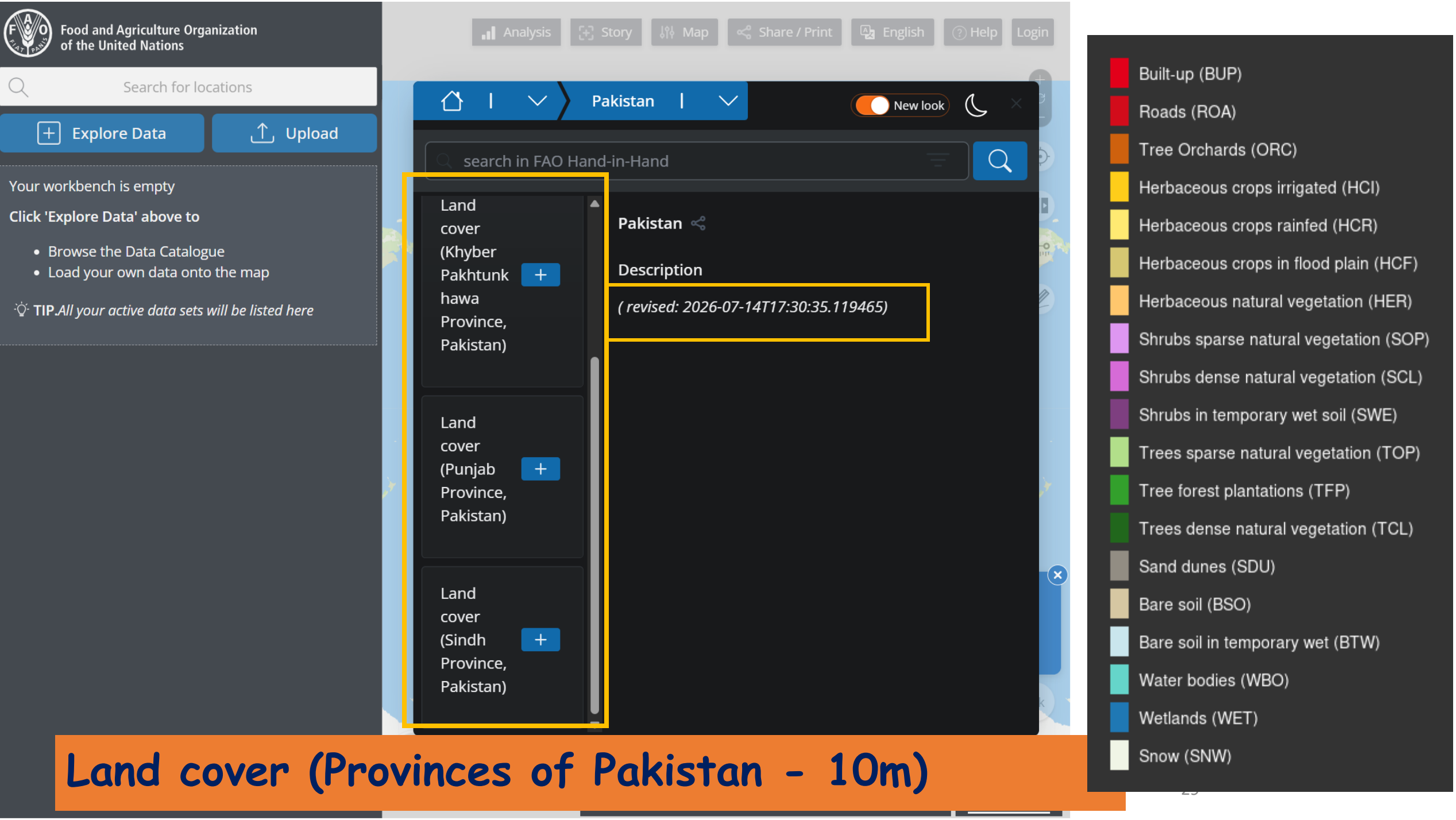
ACWA Portal Beta

GCF-FAO Project ACWA Portal combines field and remtoe sensing data.



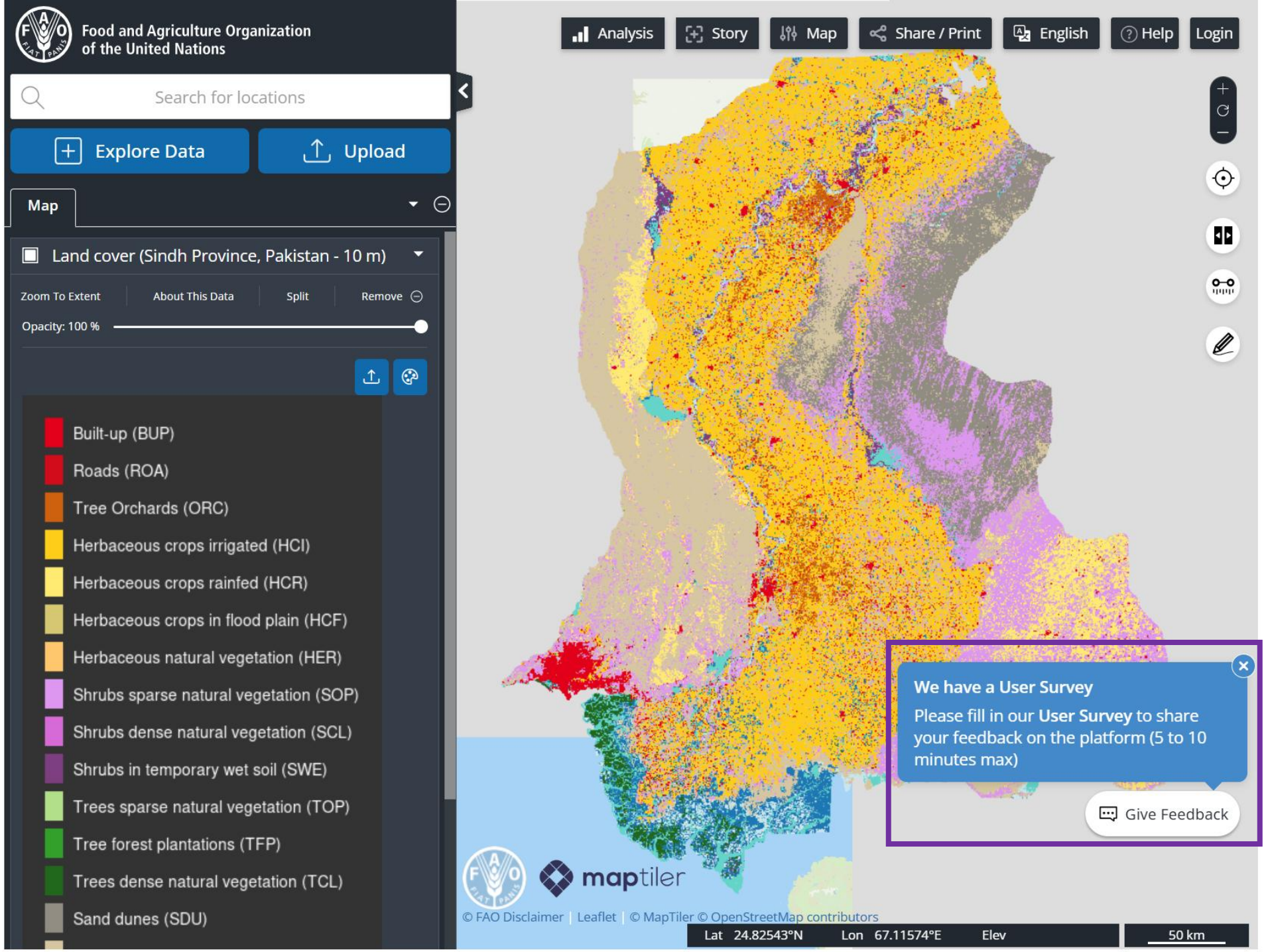




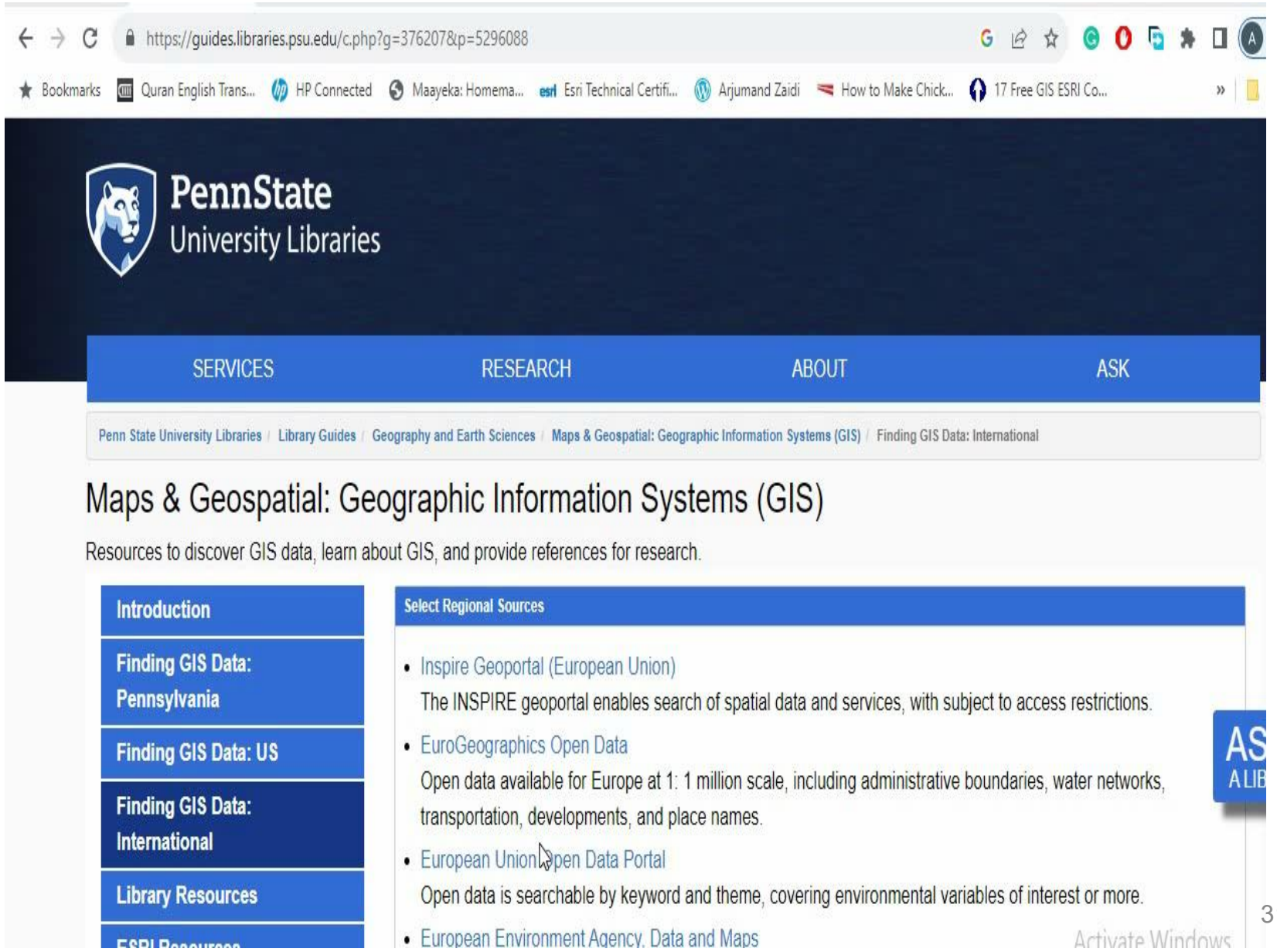


Land cover (Provinces of Pakistan - 10m)

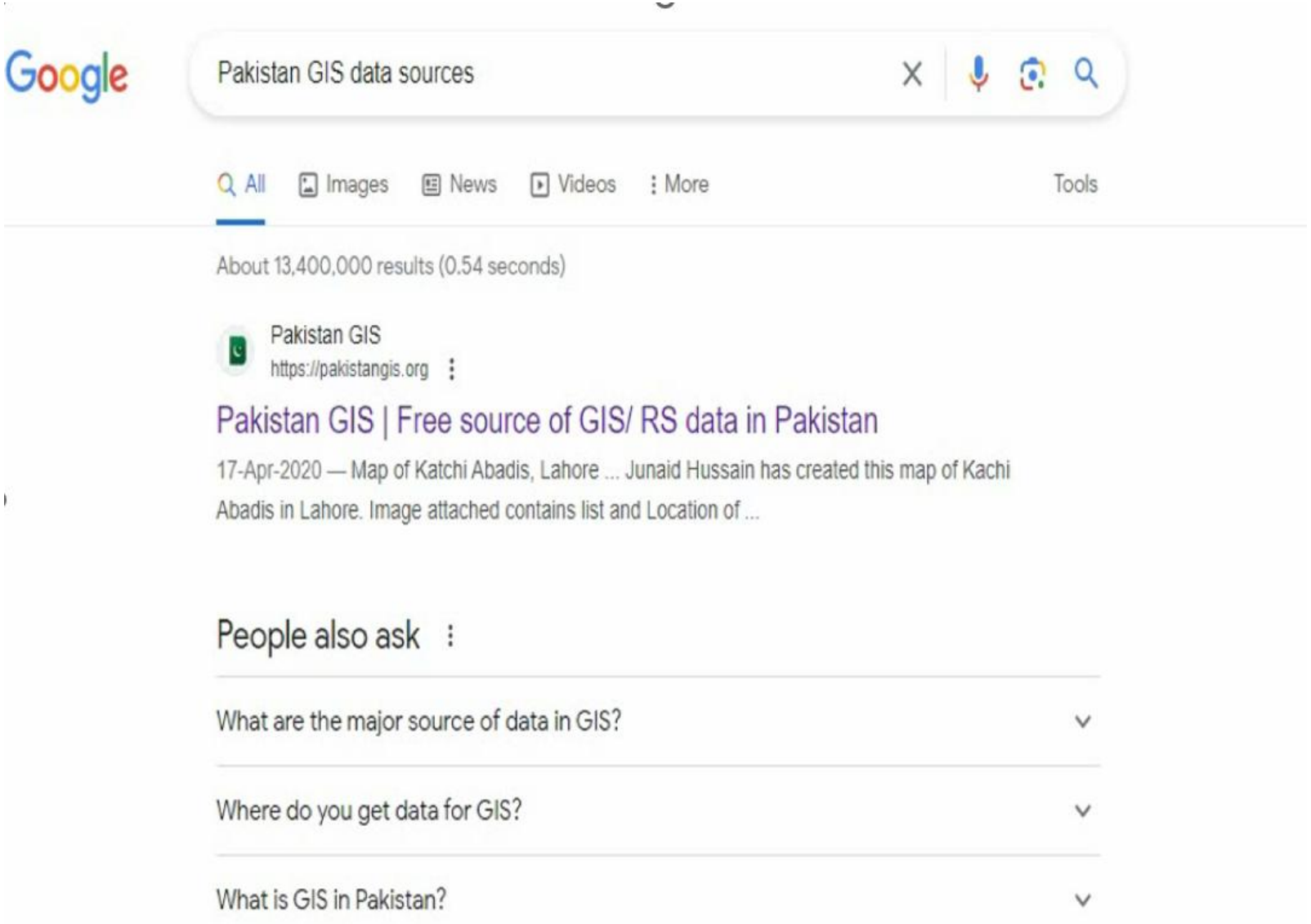
Explore this site and fill out the survey form

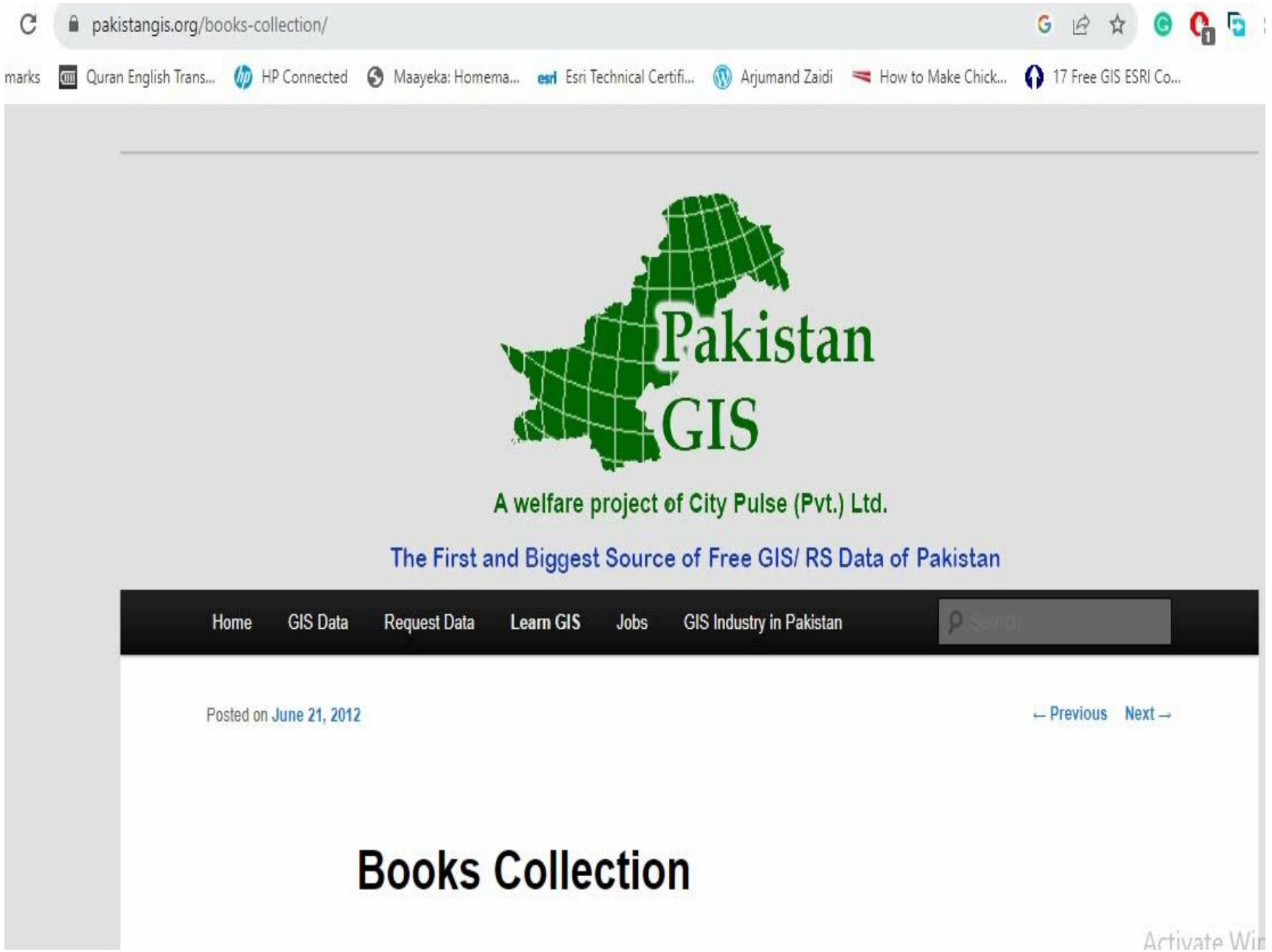


Other sources



Data For Pakistan




pakistangis.org/books-collection/
marks
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Esri Technical Certifi...
Arjumand Zaidi
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Posted on June 21, 2012
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Books Collection

Homework: Exploring Open RS and GIS Data

Objective: To identify, access, and evaluate freely available geospatial datasets for water resources applications.

Task: Search five websites that provide free RS, GIS, or hydro-meteorological datasets. Select at least one dataset from each website that could be useful for a Hydroinformatics application, such as rainfall analysis, flood mapping, drought monitoring, groundwater assessment, or irrigation management.

For details, please refer to the Homework Workbook.





Discussion/Comments/

Questions?

